

The DSD2P Questions from arXiv:1509.02061 Are Already Resolved

Math discovery status packet

August 12, 2026

Status and source questions

Status: literature already answered (three full positive answers).

In Section 4, Questions 1, 3, and 4 on PDF page 24 of [1], the authors ask:

1. whether the converse of $\text{DP} \Rightarrow \text{DSD2P}$ holds;
2. whether a Banach space X has the DSD2P if and only if X^* has the weak-star DSD2P;
3. whether $X \oplus_1 Y$ has the DSD2P whenever both X and Y do.

All three answers are yes. Kadets explicitly identifies the source questions and proves the decisive equality $\text{DSD2P} = \text{DP}$ in [2, Theorem 2.3, PDF pp. 3–4]; his introduction states that this also gives positive answers to source Questions 4.3 and 4.4. Haller–Pirk–Pöldvere had already answered the direct-sum question directly in [3, Theorem, PDF p. 1].

Identification

Kadets proves that every Banach space with the diametral strong diameter two property has the Daugavet property. The reverse implication was already known and recorded in the source. Hence

$$X \text{ has DSD2P} \iff X \text{ has the Daugavet property.} \quad (1)$$

This settles the converse of implication (1) in the source diagram.

For the dual question, use the standard dual characterization of the Daugavet property: X has the Daugavet property if and only if X^* has the weak-star Daugavet property. The weak-star analogue of Kadets’s argument (or the corresponding weak-star geometric characterization) then identifies weak-star DSD2P with weak-star Daugavet. Combining this with (1) gives

$$X \text{ has DSD2P} \iff X^* \text{ has weak-star DSD2P.}$$

Kadets explicitly records this as the positive answer to source Question 4.3.

Finally, the Daugavet property is stable under ℓ_1 -sums, so (1) implies that $X \oplus_1 Y$ has DSD2P whenever X and Y do. Independently, Haller–Pirk–Pöldvere prove exactly this assertion and explicitly say that it answers the open problem from arXiv:1509.02061.

Scope and remaining frontier

This packet resolves only the DSD2P cluster. It does *not* resolve the source questions whether DLD2P implies DD2P, whether weak-star DLD2P implies weak-star DD2P, or whether X has DD2P exactly when X^* has weak-star DD2P. The July 2026 preprint [4] still states explicitly that $\text{DLD2P} \Rightarrow \text{DD2P}$ (even $\text{DLD2P} \Rightarrow \text{D2P}$) remains open.

The bounded status check used the run indexes, the full local sources of all four cited arXiv papers, and exact-title/keyword web searches. Because Kadets explicitly cites the original paper and says he answers its Questions 4.1, 4.3, and 4.4, this belongs in `literature_already_answered`, not among new mathematical discoveries.

Human-review recommendation. Verify the question numbering on PDF page 24 of the source, Kadets's Theorem 2.3 and the paragraph immediately before it, and the direct theorem on PDF page 1 of Haller–Pirk–Pöldvere. Keep the DLD2P/DD2P and DD2P duality questions open.

References

- [1] J. Becerra Guerrero, G. López-Pérez, and A. Rueda Zoca, *Diametral diameter two properties in Banach spaces*, J. Convex Anal. 25 (2018), 817–840, arXiv:[1509.02061](#).
- [2] V. Kadets, *The diametral strong diameter 2 property of Banach spaces is the same as the Daugavet property*, arXiv:[2008.07258](#).
- [3] R. Haller, K. Pirk, and M. Pöldvere, *Diametral strong diameter two property of Banach spaces is stable under direct sums with 1-norm*, Acta Comment. Univ. Tartu. Math. 20 (2016), 101–105, arXiv:[1603.08082](#).
- [4] J. Guerrero-Viu, M. Martín, and A. Rueda Zoca, *Banach spaces with the weak diametral diameter two property*, arXiv:[2607.04370](#).